Recognizing A Pets Breed Using Deep Learning

Dataset: Oxford IIIT Pet Breed

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6147 Deep Learning Final Project — Dr. Mohammed Yousefhussien

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Why this problem:

I completed my final project on breed classification because it solves a problem a lot of pet owners actually face a lot of cats and dogs are adopted as strays or from shelters with no known background (breed) and knowing a pet's likely breed can really change how you care for it because different breeds require different exercise needs, grooming requirements, and even predispositions to certain health conditions. A lot of breeds in this dataset like Abyssinian and Bengal, or Abyssinian and Egyptian Mau look very similar especially to a person because they share the same coat pattern and similar colouring, and that is why a tool like is valuable to a new owner without breed expertise because they have no easy way to tell these apart on their own and getting it wrong means missing the breed specific care that is important.

The purpose of this model is to detect the pets breed based on a given photo. And I want to understand can a machine learning model identify what breed it is. For this I used the full Oxford IIIT Pet dataset which had 37 breeds, roughly 7349 images total I used the whole data set because I wanted the model to be able to understand all cat or dog breeds in this dataset and be able to tell them apart correctly.

To solve this problem first I built a ResNet18 based transfer learning model and fine tuned it on the full training set, then evaluated it on a test set. After that I used GradCAM to visually understand that the model was understanding characteristics of the animals rather than the background, and finally I connected the model's output to an external AI model to supply additional breed care and built a Gradio interface so the model can be used interactively.

Dataset and preprocessing:

The dataset had 37 breeds and about 200 images for breed split into a training and validation section which had 3680 images and a test section which 3669 images that torchvision data provides. I then split training and validation 85%/15% into training and validation using scikit-learn's train_test_split and set a random seed for reproducibility. After that I standardized all images in dataset using ImageNet's mean and standard deviation because the model starts from ImageNet pretrained weights and needs inputs normalized the same way as the weights were originally trained on.

The Model:

For the model I used a pretrained ResNet18 model and modified the final layer so it can classify the images into 37 different pet breeds that are in the dataset, I kept the rest of the model trainable so that it could continue learning from the Oxford pet dataset. ResNet18 already is trained on ImageNet therefore it already knows how to distinguish basic image features like shapes, edges, and textures, however finetuning it helps the model adjust its knowledge so that it can better tell the difference between pet breeds.

I trained the model using bthe SGD optimizer setting the learning rate to 0.001 and using a momentum of 0.9, also I used the StepLR scheduler that reduces learning rate by 0.1 every 7 epochs to help the model improve as training continues.

Evaluation Results:

Early stopping was used during the training process and stopped based on validation loss, this made my model stop training after 11 epochs where it reached its best validation accuracy of about 86%. On the test the model reached.

Metric	Value
Test accuracy	90.43%
Macro avg precision / recall / F1	0.91 / 0.90 / 0.90
Weighted avg precision / recall / F1	0.91 / 0.90 / 0.90

The test accuracy 90.46 was higher than the best validation accuracy at about 86% , this may be due the fact that the test images were processed in a simpler and more consistent way.

Breed performance:

From observations performance was not consistent across all breeds. Actually the strongest in distinguishing results came from breeds that have extremely distinct features;

Breed	F1 score
Keeshond	0.99
Japanese Chin	0.99
Yorkshire Terrier	0.99
Leonberger	0.98
Pug	0.98
Shiba Inu	0.97

The model had the most difficulty in telling apart similar looking cat breeds and identifying similar looking bully type dog breeds.

Breed	F1 score
American Pit Bull Terrier	0.55
Staffordshire Bull Terrier	0.58
Ragdoll	0.77
American Bulldog	0.78
Maine Coon	0.83

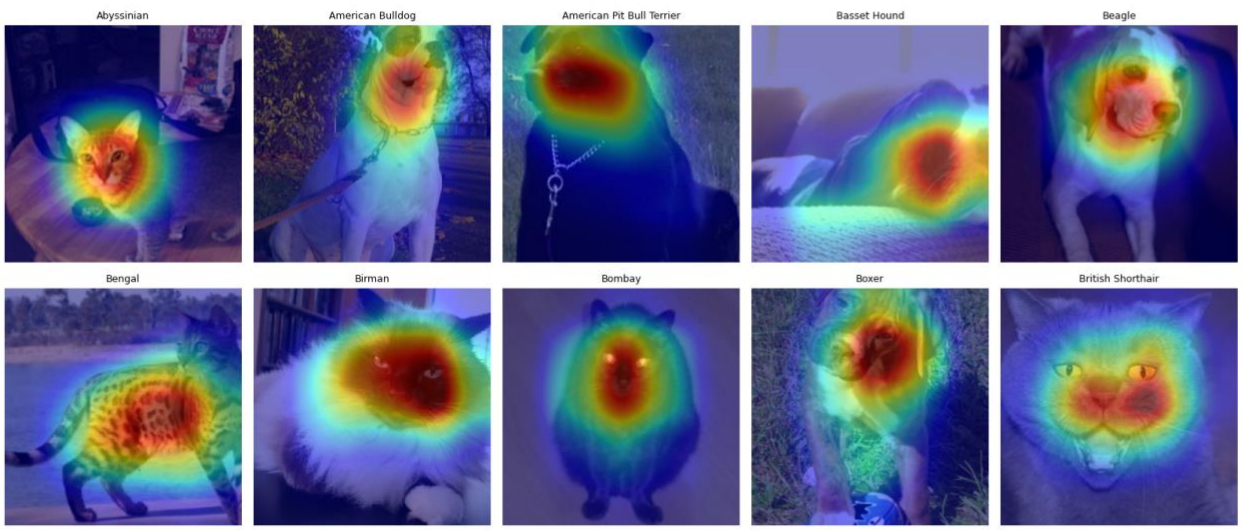
American Pit Bull Terrier and Staffordshire Bull Terrier had the lowest F1 scores out of all 37 breeds, making them the model's two weakest classes. This might be due the fact that these two breeds look very similar, especially in their body shape, head shape, and short coats. Even experienced dog handlers can have difficulty telling them apart.

The misclassification Challenge:

To better understand where the model was having a hard time, I looked at a sample of 37 mages and compared them with the incorrect predictions, and I found that the model correctly identified 30 images and misclassified 7. The errors where in similar looking breeds like the American Pit Bull Terriers, Staffordshire Bull Terriers, and American Bulldogs and for cats it confused the Ragdoll, Mainecoon, Biman, and Persian, because these breeds have similar body shapes, head shapes, and short coats, which makes them difficult to tell apart. Overall, this examination shows that most of the model's mistakes happen when breeds have very similar features. While some errors are harder to explain these results tell us that the model generally learned useful breed features but still has difficulty telling apart breeds that look alike.

Grad Cam Visualization:

I used grad cam visualization to understand what the model was looking rather than just trusting than numbers. To understand how the model was working I ran the ran cam model on a set of images and I was able to visualize the image regions that the model analyzed to make the prediction, and I was able to understand how the model was working analyzing facial, color and other features of the pets to help make the predictions and get a good accuracy.



Ai Model Integration:

An addition I made for my model was that I took a couple pictures and put them in Claud ai first to make sure that my model is working well in distinguishing and to add a little more information to the user interface for when users. I asked the ai to generate information about all the breeds that the model can detect and provide additional information that can explain to the users what breed the pet is and additional information like how big the breed can get as well as how to take care of it. I tried using a live api from an ai source to have this be done live however there was to many complications and the ai was not connecting to my model. To fix this I decided to to store the results as stored as a lookup table indexed by breed name and connected it to my gardio dashboard so that whenever a cat or dog breed is detected a command gets sent to retrieve its info from the look up table and provide the information to the users.

Interface (gardio):

After completely training the model, I built a gardio interface that is interactive with the assistance of ai. It works as follows: A user uploads a photo, and the model predicts the pet breed and shows confidence scores for all 37 breeds. Also providing AI generated breed information for the predicted breed, predictions with less than 40% confidence are have a warning mark because the model will always choose one of the 37 breeds, even if the uploaded image is not actually a pet. This makes the model's limitation clear to the user.

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Model Limitations:

The model has some limitations which can be improved with future work improving the model and adding more data to the training set like different breeds, animal types, etc. The model also has some important limitations. It can only choose from the 37 breeds it was trained on, so it still gives prediction if the image is not a pet or is a breed outside the dataset and a fix for that could be adding a function or a way to detect image not in dataset however that is too complex for the purpose of this project so it could be done in future work for now the model interface marks predictions below 40% confidence to make this limitation clear. The dataset also has only about 200 images per breed which limits how well the model can learn different poses like lighting, age, and appearances. The model can also has a hard time with unusual photos, like the upside down Abyssinian image that was wrongly predicted as a Bengal. Finally, is that the dataset only includes cats and dogs, so the model would not work for other pets such as rabbits, birds, or reptiles. A fix for all those limitations is adding more training data.

<u>Conclusion:</u>

In conclusion my goal was to build a deep learning model that is able to tell a pet's breed from a photo using the dataset Oxford-IIIT Pet dataset which has data 37 breeds. I also wanted to connect the model's prediction to I ai generated information to make the final result more useful. Overall, my goal was achieved. The final model reached a 90.43% test accuracy, with precision, recall, and F1 scores all around 0.90, using 3669 test images that was not used during training or validation. The results also showed that the model's mistakes were not completely random because It confused breeds that look very similar breeds like the American Pit Bull Terrier, Staffordshire Bull Terrier, and American Bulldog were commonly confused because they have similar body shapes, head shapes, and coats. The model also had difficulties with similar long haired cat breeds such like Persian, Birman, Maine Coon, and Ragdoll. Using all 37 breeds made it possible to identify these patterns, which may not have appeared with a smaller 10 or 20 class datasets. The use of early stoppage helped the model stop when it reached the best accuracy to prevent it from overfitting and to give us the best results. From the Grad CAM visualizations I was able to see that the model was focusing on the animals instead of the background or other unrelated parts. This might be why many of the model's mistakes happened because some breeds genuinely look similar rather than because the model was focusing on the wrong parts of the images. A limitation is that the oxford IIIT pet dataset is from a single source and has only 7349 training images. That is why the model can preform differently on photos taken with different cameras, backgrounds, lighting, or environments. However even with these limitations the model provides a strong starting point that could be improved in the future using a larger and a dataset that is varied more.